Chapter 1 Teacher Key — Energy Causes Change

> Combined answer keys: Energy Detectives · Practice · Chapter 1 Quiz.

Answer Key — Energy Detectives (Scavenger Hunt)

- **Artifact:** `production/worksheets/01-ch1-energy-detectives-student.docx`
- **Source alignment:** AP 1.1 (Energy Scavenger Hunt); Reader Ch.1 pp. 1–6; TG L1.

Item	Correct/ expected response	Acceptable alternatives	Expected pattern	Reasoning	Common misconception	Scoring note	What NOT to penalize
Part A (demo) — Before	the pencil is dull / sharpener silent	"dull pencil," "nothing happening"	states initial state	before = starting condition	—	1 pt	wording
Part A — What changed?	pencil got shorter/sharper; sharpener made noise	"pencil sharpened," "it got loud"	names a real change	change = something became different	claiming "it got hot" without evidence	1–2 pts	"I saw energy" (redirect)
Part A — Evidence	sharp tip, wood shavings, sound	any observable change	cites an observation	evidence = something observed	energy is invisible	1 pt	—
Part A — Energy form	electrical (or motion + sound)	"electricity," "motion," "sound"	names a form	sharpener runs on electricity	—	1 pt	—
Part B (hunt) — 3 rows	real changes + evidence + form	any reasonable classroom example	object/ before/after/ change/ evidence/ form filled	change is evidence of energy	"I see energy"	3 pts (1/row)	spelling
Part B — question	any reasonable question	—	a question about energy/change	curiosity about phenomena	—	1 pt	—
Part C (stored energy) — prediction	a guess (will/won't light)	any prediction	predicts before testing	prediction = testable guess	wrong prediction = failure	1 pt	a "wrong" prediction

Item	Correct/ expected response	Acceptable alternatives	Expected pattern	Reasoning	Common misconception	Scoring note	What NOT to penalize
Part C — observation	flashlight lit up	"the bulb came on"	reports observed change	light = evidence	—	1 pt	—
Part C — evidence	light appeared when switched on	"it made light"	names observed change	change proves stored energy	—	1 pt	—
Part C — explanation	the battery had stored energy; it caused change later	"battery had energy waiting"	uses stored energy concept	nonmoving object can have stored energy	only moving objects have energy	2 pts	informal phrasing

Notes

- Do not award "the battery got hot" unless the student actually observed it.
- A wrong prediction in Part C is NOT penalized if the observation/explanation are correct.

Answer Key — Chapter 1 Practice

- **Artifact:** `production/worksheets/01-ch1-energy-causes-change-practice.docx`
- **Source alignment:** Reader Ch.1 pp. 1–6; TG L1–L2; AP 1.1/2.2.

Part A — What changed?

Scenario	Observable change	Possible energy form
A light switch is flipped	room changes dark → bright	light (electrical)
A toy car is pushed	car changes still → moving	motion
A bell is rung	quiet → ringing sound	sound
An ice cube sits on a warm hand	ice melts (solid → liquid)	heat
A flashlight is turned on	bulb changes dark → glowing	light (electrical)

Part B — Evidence or not enough evidence?

Statement	Evidence?	Why
"The fan blades started spinning when we plugged it in."	YES	a change happened (still → moving)
"The battery looks shiny."	NOT enough	looking shiny is not a change caused by energy
"The room got brighter when the light came on."	YES	dark → bright is a change
"I think the motor is strong."	NOT enough	an opinion, not an observed change

Part C — Stored energy

Example	Has stored energy?	When does it cause a change?
battery in a drawer	yes	when placed in a device and used
apple held above the ground	yes	when released (it falls)
stretched rubber band	yes	when let go (it snaps/moves)
book lying flat on a table	no (no stored energy of position)	it is not raised or stretched

Part D — Explain (sample)

"Energy is the ability to cause change. When I saw the light bulb change from dark to bright, that change was evidence of energy. The stored energy in the battery was released and caused the change."

- Accept any response that (a) states energy = ability to cause change, (b) names an observed change, (c) links the change to energy as evidence.

Answer Key — Chapter 1 Quiz

- **Artifact:** `production/quizzes/01-ch1-energy-causes-change-quiz.docx`
- **Source alignment:** Reader Ch.1 pp. 1–6; TG L1–L2.

Item	Correct/expected response	Acceptable alternatives	Reasoning	Common misconception	Scoring note	What NOT to penalize
1	Energy is the ability to cause change	—	Reader p.1 definition	energy as a substance	1 pt	—

Item	Correct/expected response	Acceptable alternatives	Reasoning	Common misconception	Scoring note	What NOT to penalize
2	A change is evidence that energy is present	"a change shows energy"	Reader p.2	"energy is visible"	1 pt	wording
3	(scenario) the car changed still→moving; motion energy	"it moved; motion"	change + form	—	2 pts (1 each)	—
4	(forms) motion, sound, light, heat, electrical energy (correct option)	—	Reader p.3	—	1 pt	—
5	ENERGY → CHANGE → EVIDENCE (labels in order)	any correct chain	Reader p.2 model	swapping change/evidence	3 pts	arrow direction wording
6	(battery) the battery has stored energy released in the flashlight	"stored energy"	Reader p.4	only moving objects have energy	2 pts	—
7	A nonmoving object can have stored energy	"yes it can"	Reader pp.4–5	stored energy = moving	1 pt	—
8	(short) "Yes — the light changed from off to on, so there is evidence of energy"	any observed change + energy link	change = evidence	—	2 pts	informal phrasing
9	(evidence or not) mark which are real evidence	—	observation vs opinion	—	3 pts (1 each)	—

Out-of-scope confirmation

- No kinetic/potential/radiant/mechanical/compression-wave/inertia.
- No transfer vs transformation classification.
- No collisions or engineering design.

Total

~16 points (scored to a class-appropriate denominator; adjust per school grading policy — see OPEN_QUESTIONS).